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Professor Sabrina Senatore and Professor Zheng Yan
Editors-in-Chief
Information Sciences
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Dear Professors Senatore and Yan,

We submit our manuscript entitled "Exact Bayesian Inference for Spike-and-Slab Priors in Takagi–Sugeno–Kang Fuzzy Systems with Approximately Calibrated Model-Averaged Prediction Intervals" for consideration in *Information Sciences*.

Contribution and Novelty

This paper presents exact Bayesian inference for spike-and-slab priors in Takagi–Sugeno–Kang (TSK) fuzzy systems, delivering approximately calibrated model-averaged prediction intervals. Prior work has coupled TSK systems either with conjugate priors without sparsity, or with frequentist L_1 regularization without uncertainty. By contrast, we place a spike-and-slab prior on the rule consequents and perform posterior inference by block-Gibbs sampling with Bayesian model averaging (BMA). The method applies to any rule-based regression with linear consequents; TSK fuzzy systems are the concrete instantiation. To our knowledge, no prior work provides exact spike-and-slab posterior sampling with model-averaged prediction intervals for linear-consequent rule models; we provide it. Three results structure the contribution.

First, we provide the exact MCMC machinery: a block-Gibbs sampler whose full conditional for each rule's inclusion indicator is obtained in $\mathcal{O}(n(d+1)^2)$ time via the matrix-determinant and Woodbury identities. Its BMA predictive distribution decomposes variance into aleatoric and model-uncertainty terms by the law of total variance. This yields near-nominal 95% prediction intervals (PICP = 0.93–0.95) where our implementation of the BIC-plus-Laplace analytical pipeline undercovers (PICP = 0.00–0.18). We trace the failure to the hard-thresholding step, which collapses accuracy. In the as-implemented pipeline, a ridge-regularized fit of the ill-conditioned design matrix collapses coverage. The result is a constructive diagnosis of when analytical approximations to Bayesian model selection are safe—a question of direct interest to the uncertainty-quantification community.

Second, we document and correct a silent membership-function implementation bug in which fuzzy-set spreads are recomputed from query data at prediction time, corrupting every TSK baseline. Correcting it raises the dense TSK least-squares baseline from $R^2 = 0.48$ to 0.94 on the Energy-Cooling target. Because we reproduce this implementation pattern and quantify its effect, we release the corrected code and baselines as a reproducibility benchmark.

Third, we characterize the boundary conditions of sparsity: under correct fuzzy c-means, neither rule-level nor coefficient-level sparsity improves on the dense baseline on the low-dimensional building-energy domain. We report this as a boundary result, delineating precisely the regime in which sparsity-inducing priors justify their cost.

Journal Fit

This manuscript's core contribution is a general methodology for approximately calibrated Bayesian inference under model uncertainty—exact block-Gibbs sampling for spike-and-slab priors combined with Bayesian model averaging—with Takagi–Sugeno–Kang fuzzy systems as the concrete instantiation. The topic sits within *Information Sciences*' established treatment of uncertainty. The journal has published Bayesian approaches to uncertainty quantification (e.g., *Combining pre- and post-model information in the uncertainty quantification of non-deterministic models using an extended Bayesian melding approach*, 2019; *Bayesian approach for inconsistent information*, 2013) and counts Zadeh's generalized theory of uncertainty among its canonical references. By contrast, prediction-interval methods for fuzzy systems that remain at the application level have appeared in *Applied Soft Computing*; our contribution is methodological rather than an application of a known technique to building-energy data. We submit it as a contribution to computational intelligence and uncertainty quantification, with the fuzzy-system instantiation as the concrete example.

Manuscript Metadata

- Figures: 6 (main text)
- Tables: 1 (main text)
- References: 37
- Code and corrected baselines: `https://github.com/Jackxiaozhiren/tsk-spikeslab` (persistent DOI: 10.5281/zenodo.21929319)

Suggested Reviewers

1. Prof. Thierry Denœux, Heudiasyc, Université de Technologie de Compiègne, France, `thierry.denoeux@hds.utc.fr` — expertise in evidential and uncertainty-aware regression for fuzzy systems.
2. Prof. Tufan Kumbasar, Department of Control and Automation Engineering, Istanbul Technical University, Turkey, `kumbasart@itu.edu.tr` — expertise in type-2 fuzzy logic systems and uncertainty quantification.
3. Prof. Nikhil R. Pal, Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata, India, `nikhil@isical.ac.in` — expertise in fuzzy systems and high-dimensional TSK fuzzy modeling.

Originality and Ethical Statement

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The datasets are publicly available from the UCI Machine Learning Repository. The author declares no competing interests.

Sincerely,

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